Significant Figures and Measurement

Answer key with worked solutions

A. Understanding

1. Ruler.

> The millimetre marks make 4.3 cm certain, and you can **estimate one more digit** between the marks (4.37 cm). A fourth digit (4.372 cm) would be a guess beyond what the ruler can show.

2. Accuracy and precision.

> **Accuracy** is how close a result is to the true value; **precision** is how close repeated results are to each other. Precision describes the **spread of replicates**, so it needs at least two, and preferably three or more, measurements: one value has no spread to judge. Yes, results can be precise but not accurate: a balance that reads 0.20 g too high gives results that agree closely (small standard deviation) but are all wrong by the same amount (a systematic error).

B. Counting significant figures

3. Count.

> (a) **2** (b) **4** (c) **4** (d) **5** (e) **ambiguous**: 2, 3 or 4. Write it in scientific notation to show which, e.g. 1.20×10^{3} mL for 3.

C. Scientific notation and rounding

4. Scientific notation.

> (a) 3.50×10^{-4} m (b) 4.56×10^{4} g (c) 2.07×10^{-2} L

5. Round to 3 significant figures.

> (a) **2.46 g** (b) **0.00123 m** (c) 1.88×10^{4} **s** (18 800 s would be ambiguous) (d) **10.0 mL** (the dropped part, 51, is more than half, so 9.99 rounds up to 10.0) (e) **4.36 g** (exactly half; 6 is already even) (f) **4.38 g** (exactly half; 7 is odd, so round up to the even 8). In (c), the dropped part 50 is exactly half and 7 is odd, so 1.87 rounds to the even 1.88.

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D. Calculations

6. Multiply and divide.

> (a) 3.25 cm × 2.1 cm = 6.825 cm^2 → **6.8 cm^2** (2 s.f.) (b) 100.0 g ÷ 3.0 mL = 33.33 g/mL → **33 g/mL** (2 s.f.) (c) 8.315 J/(mol·K) × 298 K = 2477.87 J/mol → 2.48×10^3 **J/mol** (3 s.f.; the units of K cancel)

7. Add and subtract.

> (a) 12.0 g + 0.335 g + 4.5 g = 16.835 g → **16.8 g** (1 d.p.) (b) 6.38 cm − 2.1 cm = 4.28 cm → **4.3 cm** (1 d.p.)

E. Challenge

8. Aluminium block.

> $V = 2.00\ \text{cm} \times 3.00\ \text{cm} \times 4.00\ \text{cm} =$ **24.0 cm^3** (3 s.f.)
>
> $\rho = \dfrac{m}{V} = \dfrac{64.8\ \text{g}}{24.0\ \text{cm}^3} =$ **2.70 g/cm^3** (3 s.f.)

9. Triplicate weighing of a 2.500 g standard.

> (a) Mean: sum = 2.52 g + 2.48 g + 2.51 g = 7.51 g; $\bar{x} = 7.51\ \text{g} \div 3 = 2.5033\ \text{g}$ (the 3 is an exact count).
>
> Deviations: +0.0167 g, −0.0233 g, +0.0067 g. Squares: 2.778×10^{-4} g^2, 5.444×10^{-4} g^2, 0.444×10^{-4} g^2; sum = 8.667×10^{-4} g^2.
>
> $$s = \sqrt{\frac{8.667 \times 10^{-4}\ \text{g}^2}{3-1}} = 0.0208\ \text{g} \approx 0.021\ \text{g}$$
>
> $\text{RSD} = \dfrac{0.0208\ \text{g}}{2.5033\ \text{g}} \times 100\% = 0.83\%$ (the units of g cancel)
>
> (b) Error = 2.503 g − 2.500 g = **+0.003 g**; percent error = $\dfrac{+0.003\ \text{g}}{2.500\ \text{g}} \times 100\% =$ **+0.13%**
>
> (c) The error (0.003 g) is much smaller than the standard deviation (0.021 g), so the result is **accurate**: there is no sign of a systematic error. The precision is moderate (RSD 0.8%); more replicates or a more precise balance would improve it. Result: **2.50 ± 0.02 g (n = 3)**.

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